

Part A · Multiple Choice

2 points each

1. **a) Lithosphere, hydrosphere, biosphere, atmosphere, and interior**
p. 8 — the opening definition of the discipline.
2. **c) independent variable**
p. 12 — everything else must be held constant; those are the experimental controls. What changes as a result is the dependent variable.
3. **d) The Tropic of Capricorn**
p. 14 — the Equator is the only parallel that is a great circle. All other parallels, including both tropics and both polar circles, are small circles.
4. **a) the universe is expanding**
p. 24 — if every galaxy is moving away from every other galaxy, space itself must be expanding.
5. **c) an elliptical galaxy**
p. 30 — the color is the clue. Reddish and yellowish means old stars; blue means young stars still forming from gas and dust.
6. **d) during a supernova explosion**
p. 37 — fusing elements heavier than iron consumes more energy than it produces, so ordinary stars stop at iron. A supernova carries enough energy to make gold, silver, and uranium.
7. **a) Hadean, Archean, Proterozoic, Phanerozoic**
p. 45 — the first three together account for almost 90% of Earth's history.
8. **b) Superposition**
p. 47 — in an undisturbed sequence, the layers on the bottom are the oldest.
9. **b) One**
p. 49 — a ratio of 1 means 50% parent and 50% daughter, which is exactly one half-life. A ratio of 3 (25% parent, 75% daughter) would be two half-lives.
10. **d) 1:24,000**
p. 53 — the smaller the denominator, the larger the scale, and the more detail is shown. A campus map is large-scale; a world map is small-scale.
11. **b) an equal-area projection such as Gall-Peters**
p. 61 — the reader is comparing quantities spread over areas, so relative area must be true even if shapes are distorted.
12. **c) 100 feet**
p. 55 — every fifth contour line is an index contour, and $5 \times 20 \text{ ft} = 100 \text{ ft}$. Index contours are the only ones labeled.

Part B · Matching

8 points each

13. **1–c · 2–f · 3–a · 4–g · 5–h · 6–d · 7–b · 8–e**
1 point each. Cartography p. 52 · Cosmology p. 23 · Stratigraphy p. 47 · Geocoding p. 18 · Uniformitarianism p. 47 · Fossil succession p. 47 · Triangulation p. 19 · Generalization p. 52.

- 14. Relative dating establishes only the order of events — whether one rock is older or younger than another. Absolute dating assigns a numerical age in years. The discovery of radioactivity in the late 1800s made it possible.**

Rubric: 1 pt relative = order without numbers; 1 pt absolute = specific numerical age; 1 pt radioactivity (accept “radio-isotopic dating” or “isotopic dating techniques” developed in the early 20th century). (pp. 47–48)

- 15. Because a great circle must divide the Earth into two equal halves, and the Equator is the only line of latitude that does so.**

Rubric: 2 pts for the definition of a great circle — any circle that divides the Earth into two equal halves, and the largest circle that can be drawn on a sphere. 1 pt for recognizing that every other parallel is smaller and cuts the Earth unevenly, making it a small circle. (p. 14)

- 16. Remote sensing, global positioning systems (GPS), and geographic information systems (GIS).**

Rubric: 1 pt each, requiring both the name and a defensible use. **Remote sensing** — studying objects without physical contact, via satellite imagery, aerial photography, radar, or lidar. **GPS** — determining an accurate location on Earth’s surface using at least four satellites and triangulation. **GIS** — bringing spatial and non-spatial data together in layers for storage and analysis. (pp. 18–21)

- 17. A Sun-like star stops fusion and collapses into a white dwarf. A much more massive star explodes as a supernova, leaving behind either a neutron star or a black hole.**

Rubric: 1.5 pts for the Sun-like path — red giant, then fusion stops, then gravitational collapse to a white dwarf about the size of Earth, which fades out. 1.5 pts for the massive path — red supergiant, fusion continues up to iron, then core collapse produces a supernova, leaving a neutron star (core under about four solar masses) or a black hole (core over about five solar masses). (pp. 36–38)

- 18. See rubric.**

Rubric: 2 pts — why distortion is unavoidable. A globe is three-dimensional and a map is two-dimensional, so flattening always requires shearing, tearing, or compression. The orange-peel analogy earns full credit here. Since no projection can preserve every property, some distortion is guaranteed.

2 pts — the three properties. Shape, area, and distance. (Accept “shape and angle” for the first.)

2 pts — how the choice is made, with two named projections. The cartographer decides which property matters most for the map’s purpose. Expect examples such as: Mercator (conformal — correct shapes and angles, badly distorted area; good for navigation, topography, weather); Gall-Peters (equal-area — true relative sizes, distorted shapes; good for density and thematic maps); Azimuthal Equidistant (correct distance and direction from a center point; good for flight paths or earthquake epicenters); Robinson (a compromise that minimizes all distortion rather than eliminating any; good for general-purpose world maps); Goode Homolosine (interrupted — accepts tearing to preserve area).

Award full credit for any two correctly described projections tied to a purpose. (pp. 57–63)